Hands-on Exercise No. 3 DigiSkills 3.0 Batch-03 AI Using Python	Total Marks: 10 Due Date: 04/06/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
  - It is submitted after due date.
  - It is not in the required format (.doc or .docx)
  - It does not open, or file is corrupt.
  - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- Split datasets into **training and testing sets** and prepare data for Machine Learning model development.
- Train a **Linear Regression model** and generate predictions using test data.
- Evaluate and interpret Machine Learning model performance using metrics such as **MAE, MSE, RMSE, and R^2 Score**.

Problem Statement

In this exercise, you are required to build and evaluate a **Machine Learning model** using the cleaned and pre-processed dataset prepared in **Hands-on Exercise 2**. You will split the dataset into training and testing sets, train a **Linear Regression model**, and use it to predict house prices. After generating predictions, you will evaluate the model using different performance metrics such as **MAE, MSE, RMSE, and R^2 Score**. Finally, you will analyze the results and interpret whether the model performance is satisfactory or requires improvement. The objective of this exercise is to provide practical understanding of the complete workflow of a basic Machine Learning model.

You can use any programming environment such as **Jupyter Notebook, Google Colab, VS Code, or any other Python editor** to write and execute your code. After completing each task, take **screenshots of your code and its output**, and paste them into a **single MS Word (.doc/.docx) file** for submission.

Tasks:

You can download the HousePricePrediction file from LMS using the following link. (However, if you did not complete Hands-on Exercise 2, then you first need to perform the Hands-on Exercise 2 tasks on the dataset to complete the tasks of this 3rd exercise):

https://lms.digiskills.pk/Courses/AIP301/Downloads/HousePricePrediction.csv

1. Train-Test Split: **(Marks 1)**
   - Split the dataset into training and testing sets
   - Use an appropriate ratio (e.g., 80% training, 20% testing)

2. Model Training: **(Marks 2)**
   - Train a Linear Regression model using the training data

3. Model Prediction: **(Marks 2)**
   - Use the trained model to predict house prices on the test data

4. Model Evaluation; Evaluate the model using: **(Marks 2)**
   - Mean Absolute Error (MAE)
   - Mean Squared Error (MSE)
   - Root Mean Squared Error (RMSE)
   - R^2 Score

5. Interpretation: **(Marks 1)**
   - Analyze the performance of the model
   - Discuss whether the model is accurate or needs improvement

NOTE: In the solution file, no need to send/attach code file; only provide screenshots of the above tasks in MS Word File.

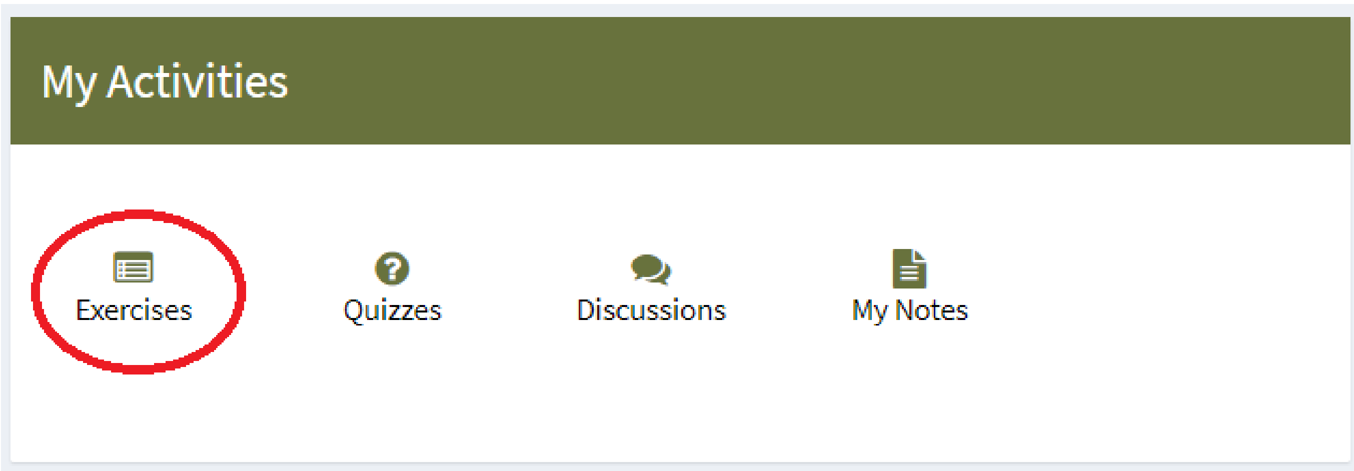
BEST OF LUCK ☺

<u>How to submit solution file on LMS?</u>

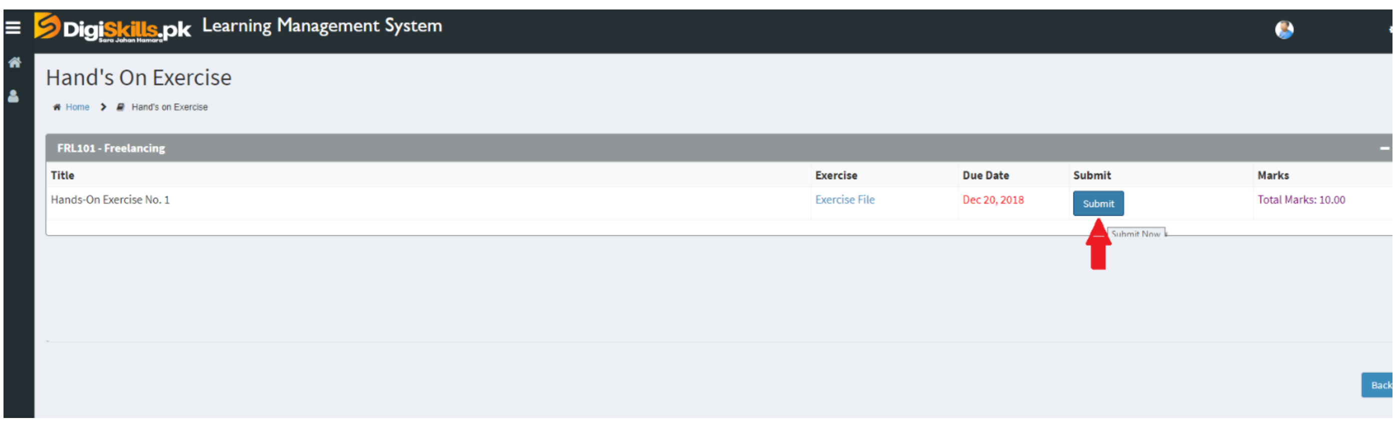
Please perform the following steps for submitting your solution using LMS:

1) Login to the LMS

2) Click on the **Exercises** button within the **My Activities** section



3) Click on the submit button to upload your Solution.



4) Keep in mind to upload your Solution in .doc or .docx format